

The ultimate remedy for personal weaknesses

Self-diagnosis · Prescribe corrective medicine · Baseafteragainchallenge · 60 minutes · 1-to-3
Online Lesson

Course type: Self-diagnostic worksheets (non-traditional handouts) - students choose their own weak points for targeted supplementation based on the simulation test data

Data source: L11 (simulation 1), L19 (simulation 2), L29 (simulation 3), L37 (simulation 4) error records of four mock tests

Core trap: ☐ T1-T9 full series - this class focuses on students' "personalized" weaknesses

How to use: ❶ Complete the "Weakness Diagnosis" → ❷ Find out the 2-3 traps where you lose the most points → ❸ Only do the practice area corresponding to the trap → ❹ Complete the "make up for the base and then challenge" test effect

Estimated number of questions: About 20-25 questions (only do your weak areas, no need to do them all)

Student Name:

Class:

Date:

Time Spent:

Part 1: My trapweakness diagnosis

Operating Instructions:

- ① Take out your four mock test papers of L11, L19, L29 and L37
- ② Count the total points lost for each trap type (you can refer to the error analysis table given to you by the teacher)
- ③ In the "Trap Diagnosis Table" below, mark the 3 traps where you **lose the most points**✓ (sorted by loss of points)
- ④ Turn to the corresponding "Top-up practice area" and only do the areas where you scored ✓
- ⑤ After completion, do the 5 questions of "Challenge after making up the bottom" for acceptance

☐ **Trap Weakness Diagnosis Table (Please ✓ next to the trap that loses the most points, select at least 1 and at most 3)**

☐ T1	carry/reverse error: The forward and backward bits are missed when adding and subtracting multiple digits, and the carry bits are superposed incorrectly in multi-digit multiplications.	☐ → Corresponds to Zone A questions 1-5
☐ T2	Decimal point misplaced: The number of decimal places in decimal multiplication products is wrong, and the decimal point in division is moved in the wrong direction.	☐ → Corresponds to Zone A questions 6-10
☐ T9	Fraction operation trap: The numerator is expanded simultaneously through division and leakage, the answer is not reduced, and improper fractions are not converted into mixed fractions.	☐ → Corresponds to Zone A questions 11-15
☐ T3	Misunderstanding of text questions: Incorrect keyword judgment ("remaining" and "more than..." are reversed)	☐ → Corresponds to Zone B questions 1-5
☐ T7	Unit conversion trap: Forgot to convert units (cm→m, mL→L, g→kg), and the units are not uniformly substituted into the formula	☐ → Corresponds to Zone B questions 6-10
☐ T4	Geometry formula confusion: The area formula is mixed up (triangle $\div 2$ is missing, trapezoid (upper base + lower base) $\div 2$ is wrong)	☐ → Corresponds to Zone C questions 1-5
☐ T5	solid geometry trap: Confusion about the concepts of volume vs. surface area, and	☐ → Corresponds to Zone C questions 6-10

✓ My choice (please fill in):

The traps where I lose the most points are: ① ☐ T____ (lose points ____) ② ☐ T____ (lose points ____) ③ ☐ T____ (lose points ____)

I will complete the following zones: ☐ Zone A ☐ Zone B ☐ Zone C (check ✓ on the areas to be done)

⚠ IMPORTANT: You don't need to do all three areas! Only work on your diagnosed areas of weakness. There are 15 questions in each area. Generally, it is enough for students to do 2 areas (about 30 questions). If time is limited, you can only do 1 area + the last 5 challenges.

The second part: Completing the basepractice area

**Zone
A**

Computational trap bottom filling (T1+T2+T9)

15 questions in total - If your diagnosis contains any of T1/T2/T9, please complete this area

A1. more number of digits advance and retreat trap T1 (question 1-5)

#	Question	Difficulty	Working Space
A1	Calculate $45678 + 29765 = ?$ (note the continuous carry)		
A2	Calculate $80003 - 45678 = ?$ (Note the zero-crossing abdication)		
A3	Calculate $1234 \times 56 = ?$ (Note the intermediate carry superposition)		
A4	Calculate $84480 \div 24 = ?$ (For long division, estimate the quotient first)		
A5	$(456 + 789) \times 12 - 3456 = ?$ (Multi-step calculation, check advance and retreat at each step)		

A2. decimal point wrong trap T2 (question 6-10)

#	Question	Difficulty	Working Space
A6	Calculate $3.14 \times 2.5 = ?$ (The decimal places of the product: $2+1=3$ digits)		

A7	Calculate $0.06 \times 0.04 = ?$ (The decimal places of the product: $2+2=4$ digits, please pay attention to padding with zeros)		
A8	Calculate $7.2 \div 0.12 = ?$ (The divisor becomes an integer: the dividend is multiplied by 100)		
A9	Calculate $0.56 \div 0.8 = ?$ (Divisor $\times 10 \rightarrow 8$, dividend $\times 10 \rightarrow 5.6$)		
A10	Compare: Are the results of 3.14×10 and $3.14 \div 0.1$ the same? Computational proof.		

A3. Fraction operation trap T9 (question 11-15)

#	Question	Difficulty	Working Space
A11	Calculate $\frac{2}{3} + \frac{1}{4} = ?$ (LCM=12, reduced after common division)		
A12	Calculate $\frac{5}{6} - \frac{1}{4} = ?$ (Note that the molecules expand simultaneously after the split)		
A13	Calculate $\frac{3}{4} \times \frac{2}{5} = ?$ (numerator $\times$ numerator, denominator $\times$ denominator, reduction)		
A14	Calculate $\frac{3}{8} \div \frac{1}{4} = ?$ ($\div$ Fraction $= \times$ Reciprocal, reduction)		
A15	Calculate $1\frac{1}{2} + 2\frac{2}{3} = ?$ (Mixed fractions $\rightarrow$ Improper fractions $\rightarrow$ Common fractions $\rightarrow$ Calculation $\rightarrow$ Restore mixed fractions)		

Zone B Text question trap supplement (T3+T7)

15 questions in total - If your diagnosis contains either T3/T7, please complete this area

B1. Text question understand trap T3 (question 1-5—must circle the key words followed by answer)

#	Question	Difficulty	Working Space (circle key word $\rightarrow$ column $\rightarrow$ calculate $\rightarrow$ answer sentence)
B1	A basketball, originally priced at \$480, is now on sale for 20% off. Xiao Ming has \$400, can he afford it? How much is the difference?		

	(Note: Comparison direction of discounted price vs ownership amount)		
B2	A ribbon is 3 meters long, made of $\frac{1}{4}$ to make a gift, and then $\frac{1}{3}$ meters to make decorations. Remaining How many meters? (Keyword: Remaining = original length – total dosage)		
B3	Xiao Ming has \$120, and Xiao Hua's money is \$35 less than 2 times Xiao Ming's SEP . How many yuan does Xiaohua have? ("Less than... times" = multiply and then subtract, don't reverse the direction). How many yuan does Xiaohua have? ("Less than... times" = multiply and then subtract, don't reverse the direction)		
B4	The warehouse originally had 50 kilograms of rice, 12.5 kilograms were shipped away in the morning, and 18.75 kilograms were shipped in in the afternoon. How many kilograms of rice are there in the warehouse now? ("Carry away" = decrease, "carry in" = increase, don't do the opposite)		
B5	A book has 240 pages. I watched $\frac{1}{3}$ on the first day, and the remaining $\frac{1}{2}$ on the second day. How many pages should I read on the third day? (The rest of the rest - two layers of reverse)		

B2. Unit conversion trap T7 (question 6-10 - if the unit is wrong, everything is wrong)

#	Question	Difficulty	Working Space
B6	A rectangle is 2.5 meters long and 150 centimeters wide. How many square meters is the area? (Convert the unit first: 150cm = ?m)		
B7	One box of juice 1.5 liters. Pour into 6 cups on average, how many milliliters per cup? (1L = 1000mL, convert to units first and then divide)		
B8	A cuboid: 0.3 meters long, 20 centimeters wide, and 15 centimeters high. What is the volume in cubic centimeters? (convert all to cm and then calculate)		
B9	A water tank can hold 2.5 liters of water. There are currently 850 ml. How many liters can be added? (Note that units are mixed: answers are in liters)		
B10	A packet of sugar weighs $\frac{1}{2}$ kg. After using 350 grams, how many grams are left? (1kg=1000g, note: if grams are used, the answer should also be in grams)		

B3. T3+T7 comprehensive (question 11-15—text question + unit conversion double trap)

#	Question	Difficulty	Working Space
B11	A rectangular garden: 12.5 meters long and 8 meters wide. To build a fence around it, \$45 per metre. How much does the fence cost in total? (Find the perimeter first, the unit is already meters. Note: the perimeter is not the area! T3+T7)		
B12	The water bottle originally contained 1.2 liters of water. After drinking 350 ml, I added $\frac{1}{4}$ liters. How many milliliters are there now? (Multiple units: liter ↔ milliliter, fraction + decimal)		
B13	The triangle has a base of 40 cm and a height of 0.3 m. What is the area in square centimeters? (Height converted to cm: 0.3m=30cm)		
B14	A rectangular container: 25 cm long, 20 cm wide, 0.15 m high. Contains $\frac{2}{3}$ water. What is the volume of water in cubic centimeters? (Find the total volume first, then $\frac{2}{3}$)		
B15	For a project, Team A did SEP , Team B did 0.35, and Team C did the rest. What fraction of the project did Team C complete? $\frac{2}{5}$, Team B did 0.35, and Team C did the rest. What fraction of the project did Team C complete?		

Zone C **Geometric trap bottom filling (T4+T5)**

15 questions in total - If your diagnosis contains either T4/T5, please complete this area

C1. Geometric formula confusion trap T4 (question 1-5—write formula in sideagain algebra)

#	Question	Difficulty	Working Space (first write formula → again algebra → calculate → answer sentence)
C1	A triangle has a base of 12 cm and a height of 8 cm. What is the area in cm ² ? (Formula: A = Bottom × Height ÷ 2 - ÷ 2 is the easiest to leak!)		
C2	A trapezoid: upper base 8 cm, lower base 12 cm, height 5 cm. What is the area in cm ² ? (Formula: A= (a+b) × h ÷ 2)		

C3	A parallelogram has a base 15 cm and a height 8 cm. What is the area in cm^2 ? (Formula: $A = \text{base} \times \text{height}$ - note: use height not hypotenuse!)		
C4	Calculation of polygon area: Divide the following figure into a rectangle (6×4) and a triangle (base 6 and height 3). Total area=? (Disassembly of composite graphics)		
C5	Compare: What is the relationship between the areas of a triangle and a parallelogram with the same base and height? Calculation proves: bottom = 10cm, height = 6cm.		

C2. 3D shape geometry trap T5 (question 6-10—do not confuse volume vs surface area)

#	Question	Difficulty	Working Space
C6	A cuboid: length 10 cm, width 6 cm, height 4 cm. Volume=? Surface area=? (Both must be counted and clearly written which is which)		
C7	A cube has side length 5 cm. Volume=? Surface area=? (Cube: $V=a^3$, $SA=6a^2$)		
C8	Drainage method: A rectangular water tank is $20\text{cm} \times 15\text{cm} \times 10\text{cm}$, and the original water height is 5cm. After placing the stone the water level rose to 7cm. What is the volume of the stone in cm^3 ? (Drainage method: $V = \text{bottom area} \times \text{water level difference}$)		
C9	The internal dimensions of a fish tank are: $30\text{cm} \times 20\text{cm} \times 25\text{cm}$. How high will the water level be after 9 liters of water are injected? (Convert the unit first: $9\text{L}=9000\text{cm}^3$, water level = volume $\div$ bottom area)		
C10	Compound solid: An L-shaped solid consists of two cuboids. A: $8 \times 5 \times 3$, B: $4 \times 5 \times 2$ (B is placed on the end above A). Total volume=?		

C3. T4+T5 comprehensive (question 11-15—area+volume double trap)

#	Question	Difficulty	Working Space
C11	A rectangular gift box: How much area is needed for the outer wrapping paper? (surface area). How much volume of items can be placed in the box? (volume). Box size: $12\text{cm} \times 8\text{cm} \times 5\text{cm}$.		
C12	The base of a triangle = the side length of a cube (6cm), and the height = the surface area of the cube $\div$ the base area. Area of triangle=? (Multi-step reasoning, T4+T5 hybrid)		
C13	A trapezoidal garden bed: upper base 5m, lower base 8m, height 4m. A layer of soil 0.3m thick should be spread on the flower bed. How many		

	cubic meters of soil are needed? (trapezoid area $\times$ thickness = volume)		
C14	The interior of a rectangular water tank: length 50cm, width 30cm, height 40cm. The original water height was 25cm. Place a rectangular iron block with a base area of 200cm ² and a height of 15cm (completely sunk). How high will the water level rise? (Advanced Drainage Method)		
C15	There is a relationship between a parallelogram and a cuboid: the base of the parallelogram = the length of the cuboid, and the height = the width of the cuboid. If the area of the parallelogram = 48cm ² , the height of the cuboid = 5cm. Volume of cuboid =? (T4+T5 cross-concept)		

The third part: Make up the base after again challenge (total 5 questions - everyone must do, test the effect of making up the base)

Challenge description:

These 5 questions mix the essence of the three areas (calculation + word questions + geometry). After completing the padding area of your choice, test your progress with these 5 questions.

Scoring method: Each question is worth 20 points, with a maximum score of 100. Answer 4 or more questions correctly = top-up is successful. Less than 3 questions answered correctly = needs further strengthening.

#	Question (Mixed T1-T9)	Difficulty	Working Space
End 1	Calculation: $\frac{3}{4} + 0.5 \times \frac{2}{3} - 0.125 = ?$ (Fractions + four decimals, T2 + T9)		
End 2	A ribbon is 5 meters long. After using SEP , I used 80 cm more for decoration. How many centimeters of ribbon are left? (Fraction + unit conversion, T3+T7+T9) $\frac{1}{4}$ Finally, another 80 cm was used for decoration. How many centimeters of ribbon are left? (Fraction + unit conversion, T3+T7+T9)		
End 3	A trapezoid: upper base 15cm, lower base 25cm, height 0.1m. What is the area in cm ² ? (Convert units first $\rightarrow$ trapezoidal formula, T4+T7)		
End 4	A rectangular water tank: 50cm $\times$ 30cm $\times$ 40cm. The original water height was 20cm. After placing an irregular stone, the water level rose to 23.5cm. What is the volume of the stone in cm ³ ? How many liters together? (Drainage method + unit conversion, T5+T7)		

End 5	Xiao Ming' s original savings is \$240. Use $\frac{1}{3}$ to buy books, and use the remaining 0.4 to buy toys. How many dollars are left at the end? (Double-layer remainder + fraction + decimal, T3+T9+T2)		
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Part 4: Self-evaluation of basebecomes results

Review of Learning Objectives - After completing this lesson you should be able to:

- ☐ Identify all trap types in our hall
- ☐ Solve  basic questions independently (100% correct)
- ☐ Challenge  Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

ultimate challengequestion number	Score (/20)	wrongreasonanalyze
End 1		
End 2		
End 3		
End 4		
End 5		
total score	/100	

Diagnosis results:

- ☐ 80-100 points: Top-up successful! Weak areas have been significantly improved.
- ☐ 60-79 points: Partial improvement. It is recommended to do the same exercises again for the wrong questions.
- ☐ Below 60 points: Weaknesses have not yet been resolved and require one-on-one follow-up by the teacher.

V. The Lessoncorecommon errorssummary

#	common error	Correct Approach
1	T1 advance and retreat bits SEP : The advance and retreat bits are missed when adding and subtracting multiple digits; the product carry in the middle of multiplication is superimposed incorrectly: The forward and backward bits are missed when adding and subtracting multiple digits; the carry bits in the intermediate product of multiplication are superimposed incorrectly	Check the advance and retreat marks at each step; the intermediate products of long multiplications must be aligned with the number of digits
2	T2 decimal point SEP : The number of decimal places in the decimal multiplication product is incorrect; the decimal point in division is moved in the wrong direction.: The number of decimal places in decimal multiplication products is calculated incorrectly; the decimal point in division is moved in the wrong direction.	Multiplication: a bit $\times$ b bit = a + b bit; division: divisor $\times 10^n$ and dividend $\times 10^n$

3	T9 Fraction SEP : The numerator is expanded simultaneously by the common fraction and leakage; the answer is not reduced; the improper fraction is not converted into mixed fractions: The numerator is expanded simultaneously through division and leakage; the answer is not reduced; improper fractions are not converted into mixed fractions	"The numerator follows the multiplication"; the last step is to check the common factors; false → bring
4	T3 text question SEP : Wrong judgment of keywords (the directions of "remainder" and "more than..." are reversed): Incorrect keyword judgment ("remaining" and "more than..." are reversed)	Circle the keywords and then draw a line segment diagram to aid understanding; write the relationship and then list it
5	T7 unit SEP : If you forget to convert the unit, substitute it into the formula; cm and m are mixed: If you forget to convert units, substitute them into the formula; use cm and m interchangeably.	All lengths must be unified in units before calculation; 1m=100cm, 1L=1000mL, 1kg=1000g
6	T4 geometric formula SEP : triangle area leaks $\div 2$; trapezoid formula is confused with parallelogram: The area of a triangle is $\div 2$; the trapezoid formula is confused with the parallelogram	First write the formula next to it → then do algebra; triangle $\div 2$, trapezoid $(a+b) \times h \div 2$
7	T5 Stereo SEP : Volume and surface area formulas are confused; drainage method water level difference is misunderstood: Confusion between volume and surface area formulas; misunderstanding of water level difference in drainage method	Volume = length $\times$ width $\times$ height; surface area = sum of the areas of all faces; drainage $V = \text{bottom area} \times \Delta h$